

Bài 1:

1a) Sử dụng biến đổi z 1 phía ta có:

$$Y'(z) + 3(z^{-1}Y'(z) + y(-1)) + 2(z^{-2}Y'(z) + z^{-1}y(-1) + y(-2)) = X'(z)$$

$$\Rightarrow Y'(z)(1 + 3z^{-1} + 2z^{-2}) + 3y(-1) + 2z^{-1}y(-1) + 2y(-2) = X'(z)$$

$$\Rightarrow Y'(z)(1 + 3z^{-1} + 2z^{-2}) + 3 + 2z^{-1} - 2 = X'(z)$$

$$\Rightarrow Y'(z)(1 + 3z^{-1} + 2z^{-2}) = X'(z) - \frac{1 - 2z^{-1}}{1 + 3z^{-1} + 2z^{-2}}$$

$$\Rightarrow Y'(z) = \frac{X'(z)}{1 + 3z^{-1} + 2z^{-2}} - \frac{1 - 2z^{-1}}{1 + 3z^{-1} + 2z^{-2}}$$

$$= \frac{1/1 + \frac{1}{3}z^{-1}}{1 + 3z^{-1} + 2z^{-2}} - \frac{1 + 2z^{-1}}{1 + 3z^{-1} + 2z^{-2}}$$

$$= \frac{1}{\frac{1 + 3z^{-1} + 2z^{-2}}{1}} + \frac{1}{\frac{1 + 3z^{-1} + 2z^{-2}}{1}}$$

$$= \frac{1}{(1 + \frac{1}{3}z^{-1})(1 + z^{-1})(2z^{-1} + 1)} - \frac{1}{1 + z^{-1}}$$

$$= \frac{1}{10} \cdot \frac{1}{1 + \frac{1}{3}z^{-1}} - \frac{3}{2} \cdot \frac{1}{1 + z^{-1}} + \frac{12}{5} \cdot \frac{1}{1 + 2z^{-1}} - \frac{1}{1 + z^{-1}}$$

$$= \frac{1}{10} \cdot \frac{1}{1 + \frac{1}{3}z^{-1}} - \frac{5}{2} \cdot \frac{1}{1 + z^{-1}} + \frac{12}{5} \cdot \frac{1}{1 + 2z^{-1}}$$

$$\Rightarrow y(n) = z^{-1} \{ Y'(z) \} = \frac{1}{10} \cdot \left(-\frac{1}{3}\right)^n u(n) - \frac{5}{2} (-1)^n u(n) + \frac{12}{5} (-2)^n u(n)$$

1b). $H(z) = \frac{1}{(1 + z^{-1})(2z^{-1} + 1)}$ có điểm cực $z = -1$ nằm trên

đường tròn đơn vị nên hệ thống không ổn định, ROC không chứa đường tròn đơn vị $\rightarrow$ không tồn tại đ/ứ tần số, đ/ứ biên độ, đ/ứ pha của hệ thống.

$$1c) \quad H(z) = \frac{1}{(1+z^{-1})(1+2z^{-1})} = \frac{A}{1+z^{-1}} + \frac{B}{1+2z^{-1}} = \frac{-1}{1+z^{-1}} + \frac{2}{1+2z^{-1}}$$

$$\Rightarrow h(n) = -(-1)^n u(n) + 2(-2)^n u(n)$$

- Hệ thống tuyến tính, bất biến, nhân quả, không ổn định, hệ thống không

$$1d) \quad E_x = \sum_{n=-\infty}^{+\infty} [x(n)]^2 = \sum_{n=0}^{+\infty} \left(-\frac{1}{3}\right)^n = \sum_{n=0}^{+\infty} \left(\frac{1}{9}\right)^n = \frac{1}{1 - \frac{1}{9}} = \frac{9}{8}$$

$$\begin{aligned} \rightarrow P_x &= 0 \\ E_y &= \sum_{n=-\infty}^{+\infty} [y(n)]^2 = \sum_{n=0}^{+\infty} \left(\frac{1}{10} \left(-\frac{1}{3}\right)^n u(n) - \frac{5}{2} (-1)^n u(n) + \frac{12}{5} (-2)^n u(n) \right)^2 \\ &= \sum_{n=0}^{\infty} \left[\frac{1}{100} \left(\frac{1}{9}\right)^n + \frac{25}{4} 1^n + \frac{144}{25} 4^n + \frac{5}{10} \left(\frac{1}{3}\right)^n + \frac{12}{25} \left(\frac{2}{3}\right)^n - 12(2)^n \right] \end{aligned}$$

$$\begin{aligned} \Rightarrow P_y &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^{N+1} \left[\frac{1}{100} \left(\frac{1}{9}\right)^n + \frac{25}{4} + \frac{144}{25} 4^n - \frac{5}{10} \left(\frac{1}{3}\right)^n + \frac{12}{25} \left(\frac{2}{3}\right)^n - 12(2)^n \right] \\ &= \frac{1}{100} \lim_{N \rightarrow \infty} \frac{1 - \frac{1}{9}}{\frac{8}{9}(2N+1)} + \frac{25}{4} \lim_{N \rightarrow \infty} \frac{N+1}{2N+1} + \frac{144}{25} \lim_{N \rightarrow \infty} \frac{1-4^{N+1}}{(1-4)(2N+1)} \\ &\quad - \frac{5}{10} \lim_{N \rightarrow \infty} \frac{1 - \left(\frac{1}{3}\right)^{N+1}}{\left(1 - \frac{1}{3}\right)(2N+1)} + \frac{12}{25} \lim_{N \rightarrow \infty} \frac{1 - \left(\frac{2}{3}\right)^{N+1}}{\left(1 - \frac{2}{3}\right)(2N+1)} - 12 \lim_{N \rightarrow \infty} \frac{1-2^{N+1}}{(1-2)(2N+1)} \end{aligned}$$

$$\Rightarrow P_y \rightarrow \infty$$

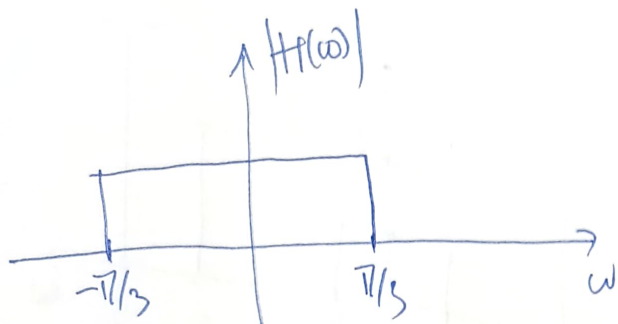
Câu 2:

$$H(\omega) = u(\omega + \pi/3) - u(\omega - \pi/3)$$

$$= \begin{cases} 1 & -\pi/3 \leq \omega \leq \pi/3 \\ 0 & \omega \neq \end{cases}$$

2a) $|H(\omega)| = \begin{cases} 1 & -\pi/3 \leq \omega \leq \pi/3 \\ 0 & \omega \neq \end{cases}$

$$\varphi_{H(\omega)} = \begin{cases} 0 & -\pi/3 \leq \omega \leq \pi/3 \\ 0 & \omega \neq \text{(hoặc không xác định)} \end{cases}$$



2b) $h(t) = \mathcal{F}^{-1}\{H(\omega)\}$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} H(\omega) e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi/3}^{\pi/3} e^{j\omega t} d\omega = \frac{1}{2\pi} \left. \frac{e^{j\omega t}}{jt} \right|_{-\pi/3}^{\pi/3}$$

$$= \frac{1}{2\pi} \frac{e^{j\frac{\pi}{3}t} - e^{-j\frac{\pi}{3}t}}{jt}$$

$$= \frac{1}{\pi t} \frac{e^{j\frac{\pi}{3}t} - e^{-j\frac{\pi}{3}t}}{2j}$$

$$= \frac{\sin \frac{\pi}{3}t}{\pi t} = \frac{1}{3} \frac{\sin(\frac{\pi}{3}t)}{\frac{\pi}{3}t}$$

$$= \frac{1}{4} \text{sinc}\left(\frac{t}{3}\right)$$

→ Hệ thống tuyến tính, bất biến, dừng, ổn định, không nhân quả.

$$2c) \quad x(t) = \sin\left(\frac{t}{2} + \frac{1}{3}\right) + 2\cos\left(2t - \frac{1}{2}\right) + 3\sin(3t)$$

$$= \frac{1}{2j} \left(e^{j\frac{t}{2} + j\frac{1}{3}} - e^{-j\frac{t}{2} - j\frac{1}{3}} \right) + e^{j2t - j\frac{1}{2}} + e^{-j2t + j\frac{1}{2}}$$

$$= \frac{1}{2j} \left(e^{j\frac{t}{2} + j\frac{1}{3}} - e^{-j\frac{t}{2} - j\frac{1}{3}} \right) + \frac{3}{2j} \left(e^{j3t} - e^{-j3t} \right)$$

$$C_{1/2} = \frac{e^{j/3}}{2j} = \frac{1}{2} e^{j(\pi/2 - 1/3)}$$

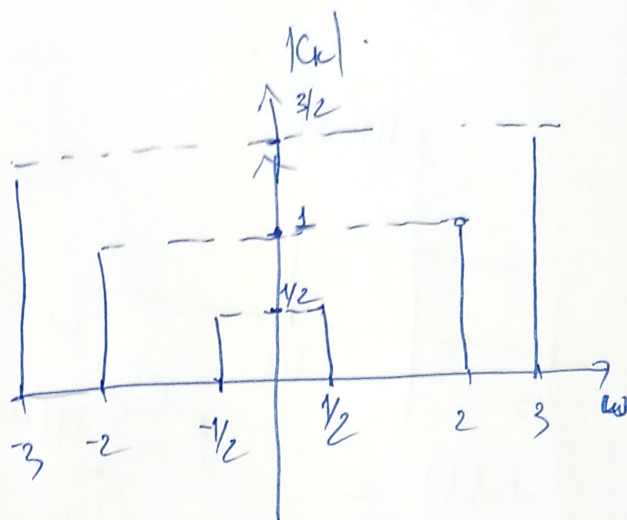
$$C_{-1/2} = -\frac{e^{-j/3}}{2j} = \frac{1}{2} e^{j(\pi/2 - 1/3)}$$

$$C_2 = e^{-j/2}$$

$$C_{-2} = e^{j/2}$$

$$C_3 = \frac{3}{2j} = \frac{3}{2} e^{-j\pi/2}$$

$$C_{-3} = -\frac{3}{2j} = \frac{3}{2} e^{j\pi/2}$$



$$|C_{1/2}| = |C_{-1/2}| = \frac{1}{2}$$

$$|C_2| = |C_{-2}| = 1$$

$$|C_3| = |C_{-3}| = \frac{3}{2}$$

$$\varphi_{C_{1/2}} = -\pi/2 + \frac{1}{3}$$

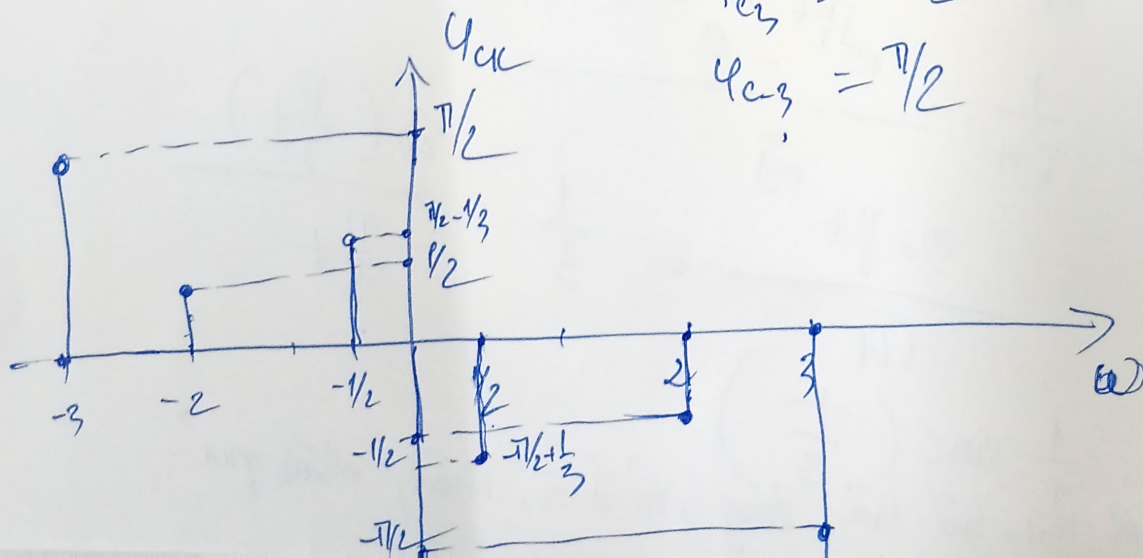
$$\varphi_{C_{-1/2}} = \pi/2 - \frac{1}{3}$$

$$\varphi_{C_2} = -\frac{1}{2}$$

$$\varphi_{C_{-2}} = \frac{1}{2}$$

$$\varphi_{C_3} = -\pi/2$$

$$\varphi_{C_{-3}} = \pi/2$$



2d:

$$y(t) = C_{1/2} e^{jt/2} H\left(\frac{1}{2}\right) + C_{-1/2} e^{-jt/2} H\left(-\frac{1}{2}\right) + C_2 e^{j2t} H(2) + C_{-2} e^{-j2t} H(-2) + C_3 e^{j3t} H(3) + C_{-3} e^{-j3t} H(-3)$$

$\Rightarrow w = 2, -2, 3, -3 \notin \left[-\frac{\pi}{3}, \frac{\pi}{3}\right]$

$\Rightarrow H(w) = 0 \mid w = 2, -2, 3, -3$

$\Rightarrow y(t) = C_{1/2} e^{jt/2} + C_{-1/2} e^{-jt/2} = \sin\left(t/2 + 1/3\right)$

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